Reminder:

- The *expected value* of a random variable is the long run average value. For example, the expected outcome when rolling a fair die is 3.5 as that is the likely long run average roll.

  The formula for the expected value of a random variable X is given by:
  $$\mathbb{E}(X) = \sum_r r \times \mathbb{P}(X = r)$$
  where the sum is taken over all possible values of X.

New idea:

- The *variance* of a random variable is the expected value of the squared deviation of outcomes from the expected value itself, given by the formula below:
  $$Var(X) = \mathbb{E}\left[\left(X - \mathbb{E}(X)\right)^2\right]$$

  This is potentially difficult to work out and so the below formula is usually easier to use:
  $$\mathbb{E}(X^2) - (\mathbb{E}(X))^2$$

1. The random variable X has a probability distribution given by
   $$P(X = x) = kx^2 - x, \qquad \text{for } x = 1, 2, 3, 4, 5.$$
   a. Complete the following table:

x	1	2	3	4	5
$P(X = x)$	$k - 1$	$4k - 2$			

   b. Find the value of k.
   c. Find $E(X)$ and $Var(X)$.

2. Two fair die are rolled. Let S model the smallest number appearing on the two dice.
   a. Find the probability distribution for S.
   b. Find $E(S)$ and $Var(S)$.

3. Consider the following bet in a game of roulette. £1 on 20 (a black number), £3 on black, and £1 on zero. Let P model the profit on this game (so that $P = -5$ if all bets lose).
   a. Find the probability distribution for P.
   b. Find $E(P)$ and $Var(P)$.

4. a) A man takes part in a game in which he throws two fair dice and scores the sum of the two numbers shown. The rewards for the scores are in the following table:

Score	12	10	7	5	Other
Reward (£)	16	6	3	5	0

   Calculate the expected reward for throwing two dice.

   b) A bag contains five identical disks of which two are black and the others are white. The disks are drawn without replacement until both black disks have been drawn. You win an amount of money equal to the number of draws squared. Find the expectation and variance of the random variable X, defined to be the amount you win.

5. There are two versions of a gambling game. In version 1 the player flips a fair coin three times, and wins 1 for each head obtained. Let X denote the amount won when playing this version. In version 2 a pack of six cards contains three cards labelled H, and three cards labelled T. The player draws three cards at random without replacement from the pack, and wins 1 for each H obtained. Let Y denote the amount won when playing this version.
   a. Find the probability distributions of X and Y.
   b. Show that the expectation of X is equal to that of Y. If this gambling game is 'fair', then what should the entrance fee be?
   c. Find Var(X) and Var(Y).
   d. Which version of this game would you prefer to play? Give reasons for your answer.

6. A gambler uses the following strategy to play roulette: he puts £1 on red. If he wins he is up £1. If he loses then he puts £2 on red. If he wins now then he has spent $1 + 2 = £3$, and won £4, so overall he is up £1. If he loses then he puts £4 on red. If he wins now then he has spent $1 + 2 + 4 = £7$ and won £8 so he is up £1. If he loses then he puts £8 on red, and so on.

   It might seem like the gambler has a no-lose strategy, but in practice gambling machines/casinos have maximum bets, and the gambler has a limit to the amount of money he can spend. For the purposes of this question, let's assume that the maximum bet the gambler is allowed to place is £512 (or equivalently that he has a maximum of £1023 to spend). Therefore his strategy ends either with a win, in which case he wins £1, or a catastrophic failure, in which case he loses £1023.
   a. Check that you understand how the two outcomes describe above arise, and find the probability of each of them occurring.
   b. If P models the profit made by the gambler, find $E(P)$ and $Var(P)$.

7. Let M model the middle score when three dice are rolled.
   Find the expectation and variance of M.

8. **BONUS:** Suppose someone with exactly £16,383 pounds (one less than a power of 2) takes all their money to a casino with a no-limits roulette table, and follows the strategy described in question 5. Suppose also that they continue to apply the strategy (winning £1 or losing it all) until they have either doubled their money, or lost their original money (and are left with only what they've won up to that point.)
   Find the probability that they are successful, and the expected amount of money they are left with [hard].